



Current class diagrams of the game

This document describes the classes and runtime boundaries inferred from the current source code in `include/` and `src/`. The diagrams are split by responsibility so each slice can be read without confusing ownership relations with usage relations.

Scope and how to read

- `*--` is composition: the left object owns the component's lifetime (usually a by-value member or a `std::unique_ptr`).
- `o--` is aggregation/non-owning recording; `-->` and `..>` are references, calls, or return types — they transfer no ownership.
- Cardinality labels (`1` , `0..1` , `0..*` , `2`) reflect the runtime members/collections, not the number of statically created C++ classes.
`<<RAII token>>` is a role declared or shown directly in the code.
`SaveManager` deliberately carries no Singleton label.
- Class, enum, and method names stay in English as in C++; the explanations are prose. Long templates are shortened inside class boxes to keep Mermaid stable; the precise owning type is recorded in the notes and evidence table.

The color scheme is consistent across blocks: orange = core/owner, blue = state or physics, green = gameplay/entity, purple = pattern/event, and light red = enemy. Colors are only a visual aid; semantics are decided by the arrows and notes.

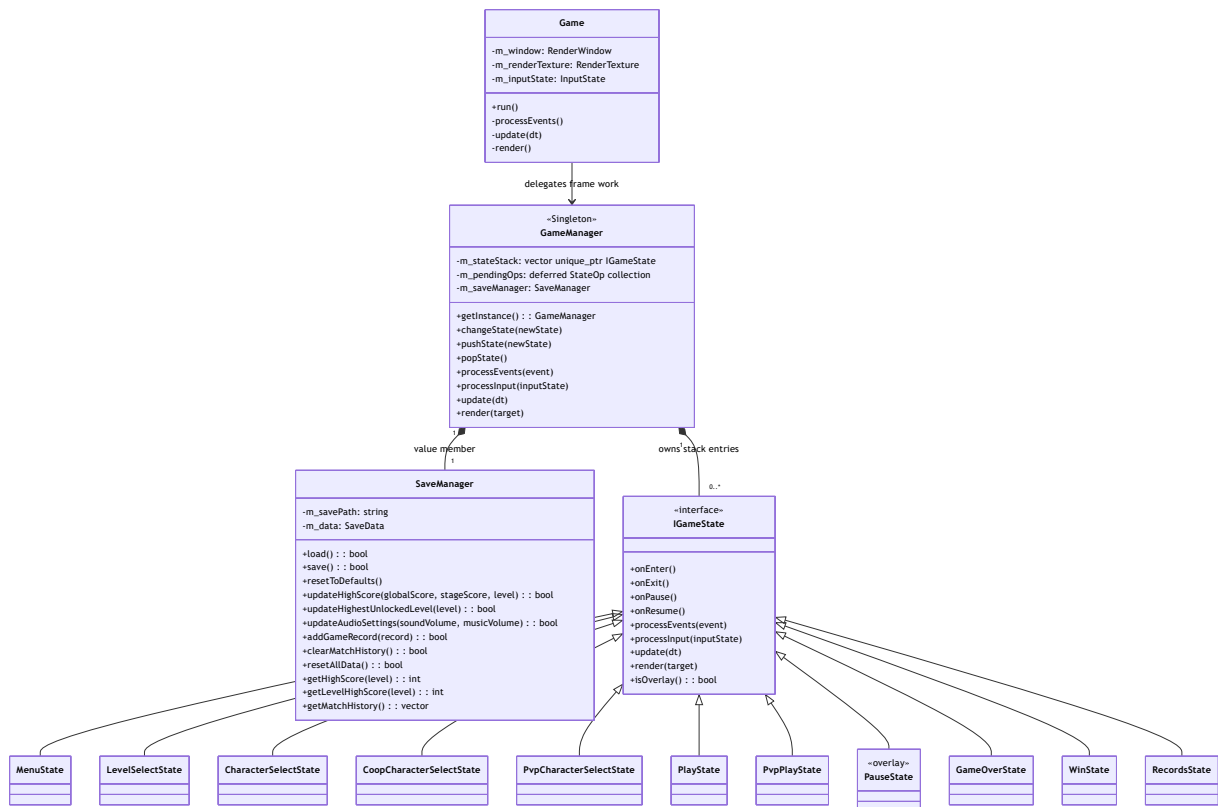
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1. Application loop and state stack

This is the path of one frame: Game receives events and keeps the InputState, then delegates input, update, and render to GameManager. GameManager is a Singleton coordinating a stack of unique_ptr<IGameState>; SaveManager is merely a by-value member of the manager, not an independent Singleton.

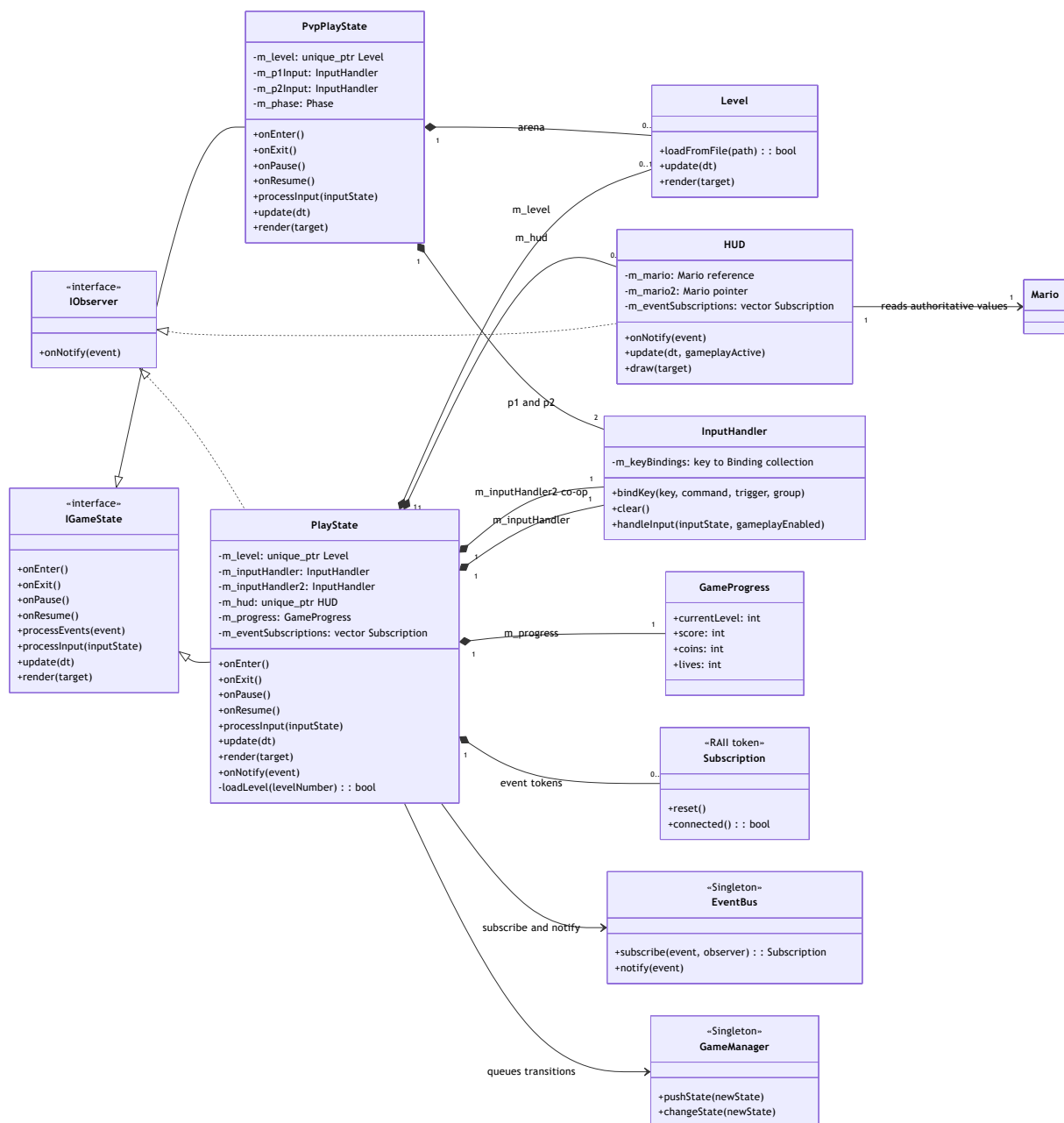


GameManager::changeState, pushState, and popState only enqueue a PendingOp; only at the end of GameManager::update() does processPendingOps() run. CHANGE calls onExit() on the old stack then onEnter() on the new state; PUSH calls onPause() on the state below and

POP calls `onResume()` on the new top state. When the top state is an overlay, `render()` draws the state below first. These behaviors are real lifecycle semantics, not just inheritance relations on the picture.

2. Ownership boundary of the gameplay states

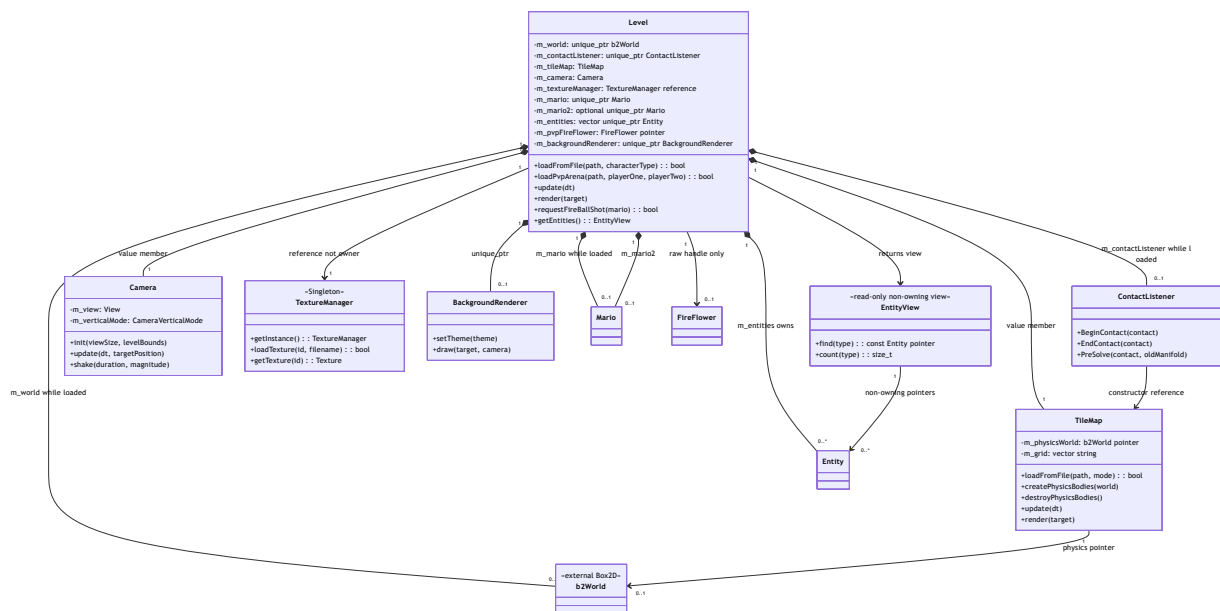
`PlayState` is the only campaign state owning the current level, HUD, two by-value `InputHandler`s (the second handler is for co-op), session progress, and event subscription tokens. `PvpPlayState` has an arena `Level` and two input handlers. Neither state owns `Mario` directly; `Level` is the owner.



In `PlayState::onEnter()` , four subscriptions are kept in `m_eventSubscriptions` ; `onExit()` releases them before stopping the music. On receiving `GAME_PAUSED` , the state enqueues `PauseState` through `GameManager` instead of owning the overlay itself. `HUD` holds a `const Mario&` and optionally a second `const Mario*` , so the arrow to `Mario` means reading data, not ownership.

3. Level, physics, and the entity collection

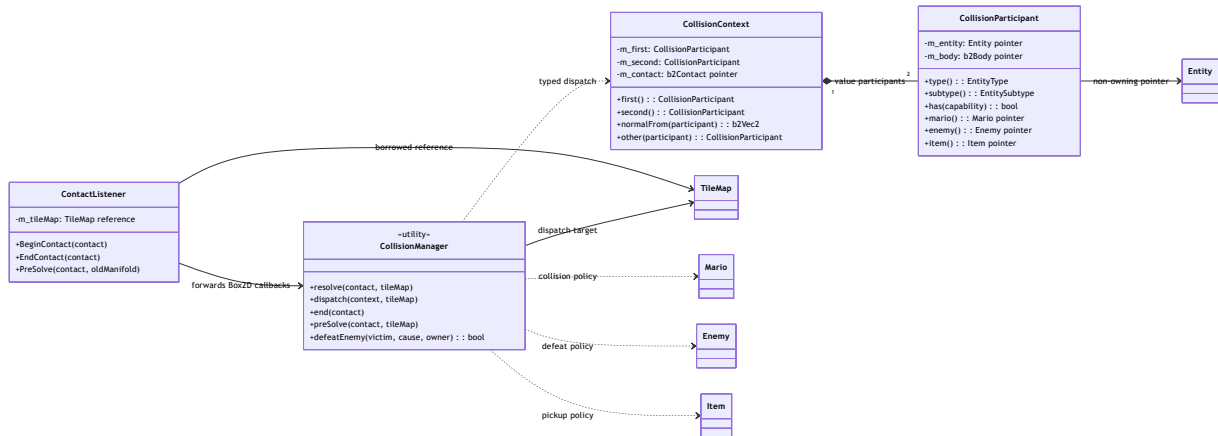
`Level` is the ownership boundary of a stage: the `Box2D` world and listener are `unique_ptr` s, `TileMap` / `Camera` are by-value members, the primary and second `Mario` (co-op/PvP) are `unique_ptr` s, and spawned entities are gathered in a `vector<unique_ptr<Entity>>` . `TextureManager&` is a reference taken from the global Singleton; `Level` does not release it. `m_pvpFireFlower` is only a raw handle to a flower already inside `m_entities` .



`Level::spawnEntitiesFromTileMap()` creates a local `EntityFactory` , calls `create(SpawnRequest, SpawnContext)` , attaches the `TextureManager` , and pushes the result into `m_entities` ; `Mario` is created separately beforehand. Therefore the `Level – Entity` relation in the picture is true ownership, while `getEntities()` only returns a view that expires when `Level` loads/updates/mutates.

4. Collision callback boundary

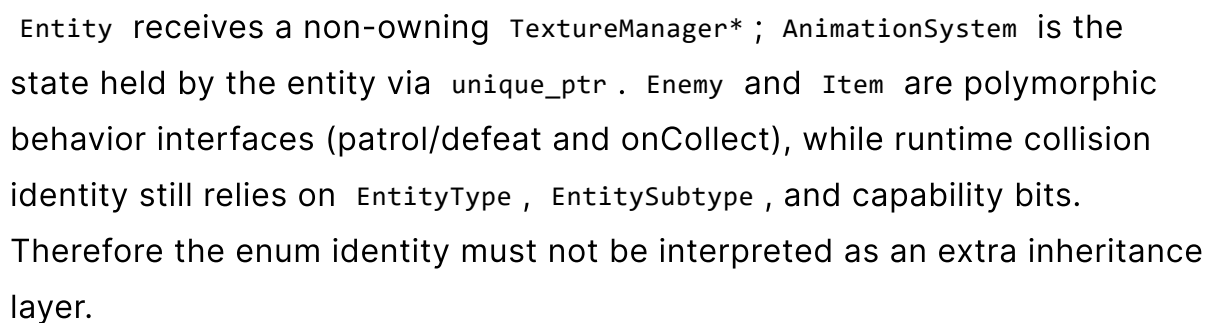
Box2D calls `ContactListener` ; the listener forwards the callback to `CollisionManager` . This utility creates/receives a `CollisionContext` containing two type-checked `CollisionParticipant` s. The participants merely point at live entities and bodies; they never change Level's ownership.



This is dependency/callback, not a new object graph: `ContactListener` 's constructor takes a `TileMap&` ; `CollisionManager` has a deleted constructor and only provides static helpers. Compatibility overloads like `isMario()` or `isItem()` on `Entity` are not drawn as new subclasses.

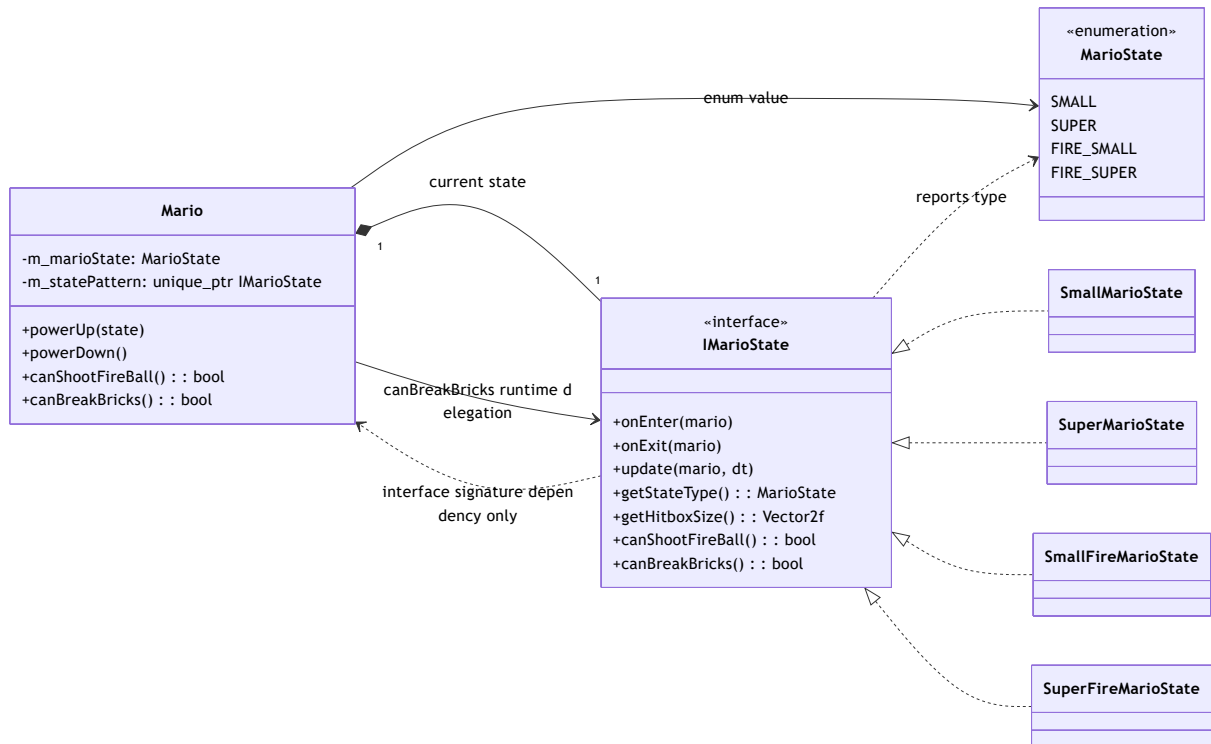
5. Entity inheritance tree

This diagram separates the hierarchy from Level's ownership. The enemy and item branches list the current concrete headers; other direct world objects such as `Elevator` , `Springboard` , `Toad` , `ScorePopup` , `BlockDebris` , `BowserAxe` , and `BulletBillLauncher` are still `Entity` s but are omitted from the enemy/item slice.



6. Mario power-up state

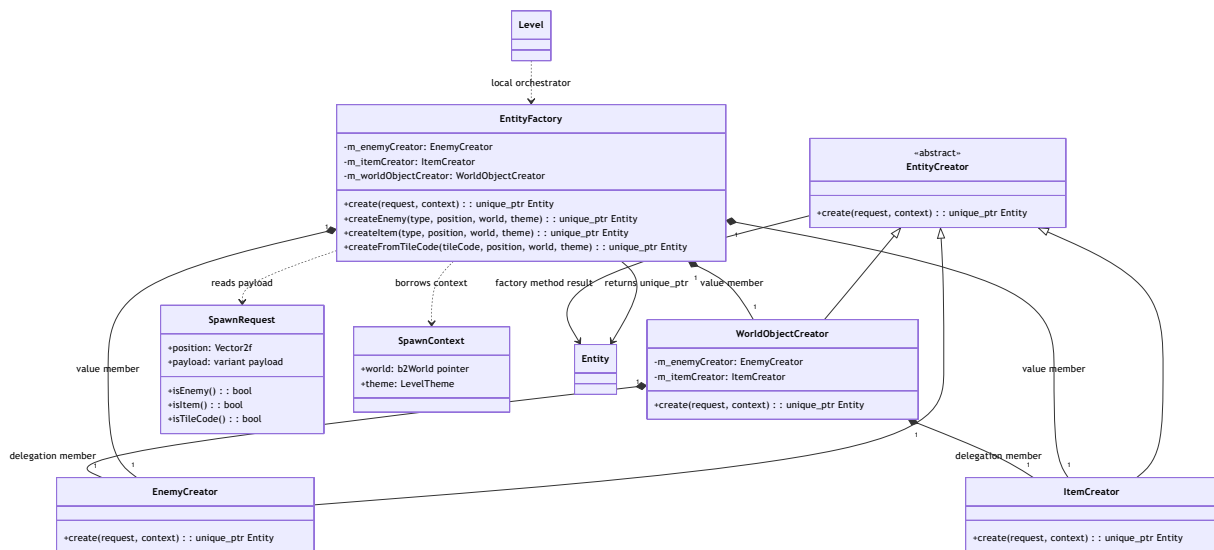
Besides the game's state stack, Mario has its own State Pattern. Mario holds the current state in a `std::unique_ptr<IMarioState>` and swaps the implementation when `applyStateTransition()` changes `MarioState`; the four concrete states only supply the hitbox, fireball, and brick-breaking rules.



These are two different state seams: `IGameState` controls the lifecycle of game screens; `IMarioState` controls the power-up rules of one Mario. The methods `onEnter`, `onExit`, `update`, `getHitboxSize`, and `canShootFireBall` remain interface signatures, but there is no production call through `m_statePattern` for them in the current `Mario.cpp`. The dashed arrow to `Mario` is therefore only a parameter-type dependency; the single runtime arrow in this slice is `Mario::canBreakBricks()` calling `m_statePattern->canBreakBricks()`. Do not merge the two state seams into one shared hierarchy, and do not assign `saveManager` to the State Pattern.

7. Factory Method: request -> creator -> entity

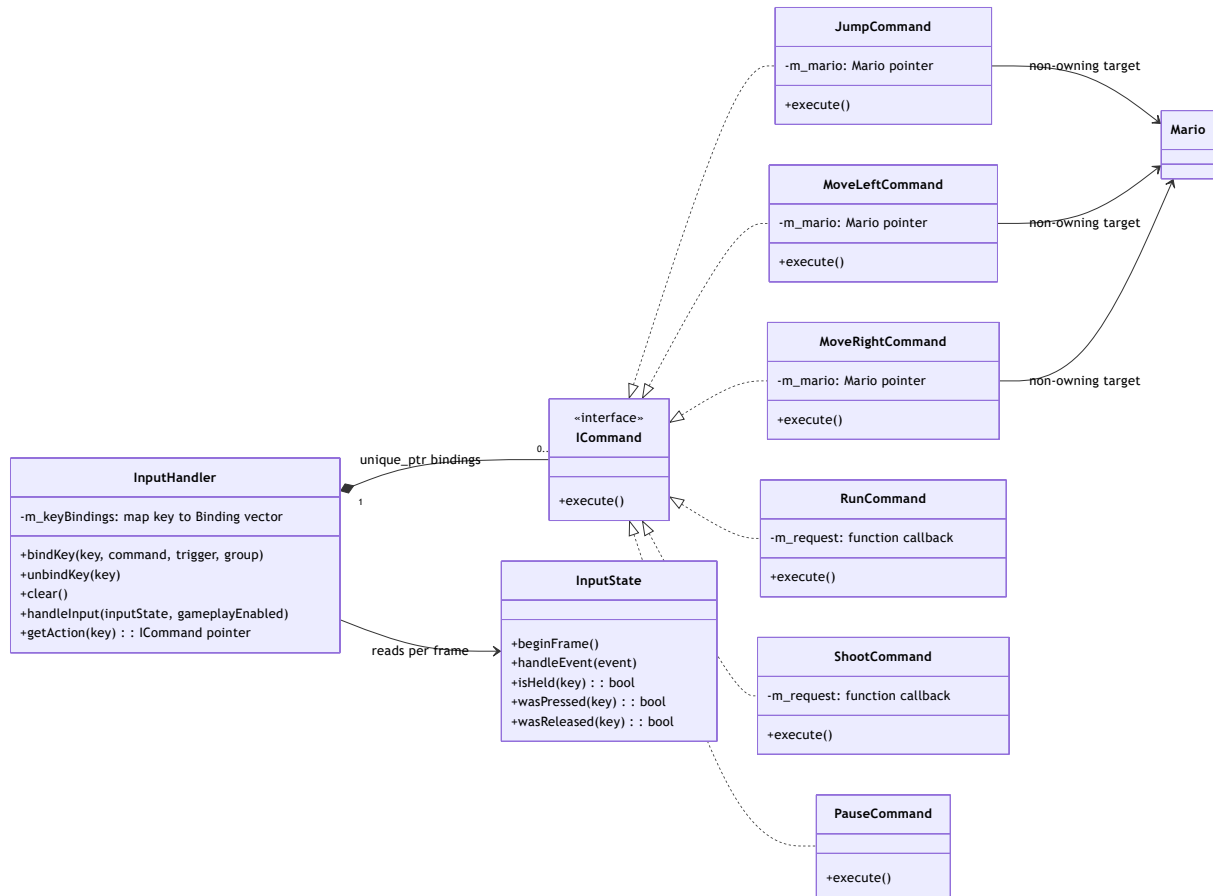
`EntityFactory::create()` is the canonical entry point. A `SpawnRequest` carries exactly one variant payload among `EnemyType`, `ItemType`, or a tile code; `SpawnContext` passes only a non-owning `b2World*` and the theme. `EntityFactory` holds three creators by value; `WorldObjectCreator` in turn holds two stateless creators by value. There is no Singleton label on the factory: the old static helpers merely forward to the default orchestrator.



In `Level::spawnEntitiesFromTileMap()`, the factory creates the object and only then does `Level` attach the texture and tile map for enemies and move the `unique_ptr` into the collection. Only `Level` receives ownership of the result; creators keep no entity after returning it.

8. EventBus and Observer with the RAII token

`EventBus` implements `ISubject` and is a Singleton. `subscribe()` returns a move-only `Subscription`; the token holds a `shared_ptr<SubscriptionLease>`, while `EventBus` state stores only weak lease records. The `IObserver*` inside a lease is a non-owning pointer. When the last token is destroyed or `reset()`, the registration is disconnected.



`PlayState::rebindCommands()` builds the concrete commands with `m_level->getMario()` and maps the shoot callback to `Level::requestFireBallShot()`. This is a dependency created at the call site, not field ownership from the command to Level.

10. Source evidence and representation limits

Diagram relation/contract	Current evidence
Game loop delegates to the manager	_include/core/Game.h:14 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/Game.h#L14) _src/core/Game.cpp:78 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L78) _src/core/Game.cpp:105 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L105) _src/core/Game.cpp:129 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp#L129)
Singleton, deferred state stack, value SaveManager	_include/core/GameManager.h:23 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L23) _include/core/GameManager.h:72 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L72) _include/core/GameManager.h:74 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/GameManager.h#L74) _include/core/SaveManager.h:22 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SaveManager.h#L22) _src/core/GameManager.cpp:27 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L27) _src/core/GameManager.cpp:76 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L76) _src/core/GameManager.cpp:88 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L88)

Diagram relation/contract	Current evidence
change/push/pop lifecycle and overlay rendering	_src/core/GameManager.cpp:44 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L44) _src/core/GameManager.cpp:57 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L57) _src/core/GameManager.cpp:64 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L64) _src/core/GameManager.cpp:109 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/GameManager.cpp#L109)
Concrete game states and the state contract	include/states/IGameState.h:19 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/IGameState.h#L19) include/states/PlayState.h:23 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h#L23) include/states/PvpPlayState.h:24 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PvpPlayState.h#L24) include/states/PauseState.h:13 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PauseState.h#L13)

Diagram relation/contract	Current evidence
PlayState owns Level/HUD/handlers/tokens	<u>include/states/PlayState.h:77</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h:77) <u>include/states/PlayState.h:78</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h:78) <u>include/states/PlayState.h:82</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h:82) <u>include/states/PlayState.h:106</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/PlayState.h:106) <u>include/ui/HUD.h:37</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/ui/HUD.h#L37) <u>src/states/PlayState.cpp:281</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/states/PlayState.cpp:281)

Diagram relation/contract	Current evidence
Level ownership and non-owning handles	_include/level/Level.h:30 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L30) _include/level/Level.h:210 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L210) _include/level/Level.h:211 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L211) _include/level/Level.h:212 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L212) _include/level/Level.h:215 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L215) _include/level/Level.h:218 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/include/level/level.h#L218) _src/level/Level.cpp:174 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L174) _src/level/Level.cpp:175 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L175)
Level factory call and entity adoption	_src/level/Level.cpp:685 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L685) _src/level/Level.cpp:705 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L705) _src/level/Level.cpp:730 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L730) _src/level/Level.cpp:737 (https://github.com/baoduo/Group04-FinalProject/blob/main/05_Source_Code/src/level/Level.cpp#L737)

Diagram relation/contract	Current evidence
Entity/Character/Mario/Enemy hierarchy	<u>include/entities/Entity.h:31</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Entity.h:31) <u>include/entities/Character.h:17</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Character.h:17) <u>include/entities/Mario.h:54</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Mario.h:54) <u>include/entities/Enemy.h:21</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Enemy.h:21) <u>include/entities/Goomba.h:17</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Goomba.h:17) <u>include/entities/Koopa.h:26</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Koopa.h:26) <u>include/entities/PiranhaPlant.h:15</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/PiranhaPlant.h:15) <u>include/entities/BuzzyBeetle.h:13</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/BuzzyBeetle.h:13)

Diagram relation/contract	Current evidence
Item hierarchy and collection contract	_include/items/Item.h:14 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/Item.h#L14) _include/items/Item.h:26 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/Item.h#L26) _include/items/Coin.h:18 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/Coin.h#L18) _include/items/Mushroom.h:19 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/Mushroom.h#L19) _include/items/FireFlower.h:12 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/FireFlower.h#L12) _include/items/Star.h:13 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/items/Star.h#L13)
Mario power-up State Pattern	_include/states/IMarioState.h:17 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/states/IMarioState.h#L17) _include/entities/Mario.h:205 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/entities/Mario.h#L205) _src/entities/Mario.cpp:252 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/entities/Mario.cpp#L252) _src/entities/Mario.cpp:963 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/entities/Mario.cpp#L963) _src/entities/Mario.cpp:1486 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/entities/Mario.cpp#L1486) _src/entities/Mario.cpp:1490 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/entities/Mario.cpp#L1490)

Diagram relation/contract	Current evidence
Read-only EntityView lookup results	<u>include/level/EntityView.h:25</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/EntityView.h:25) <u>include/level/EntityView.h:133</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/EntityView.h:133) <u>include/level/EntityView.h:143</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/EntityView.h:143) <u>include/level/EntityView.h:156</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/EntityView.h:156) <u>include/level/EntityView.h:171</u> (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/level/EntityView.h:171)

Diagram relation/contract	Current evidence
Collision callback and typed context	include/physics/ContactListener.h:14 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/physics/ContactListener.h#L14) include/physics/CollisionManager.h:33 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/physics/CollisionManager.h#L33) include/physics/CollisionManager.h:61 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/physics/CollisionManager.h#L61) include/physics/CollisionManager.h:95 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/physics/CollisionManager.h#L95) src/physics/ContactListener.cpp:10 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/physics/ContactListener.cpp#L10) src/physics/ContactListener.cpp:14 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/physics/ContactListener.cpp#L14)

Diagram relation/contract	Current evidence
Factory Method creator composition	<u>include/patterns/EntityFactory.h:20</u> <u>include/patterns/EntityFactory.h:48</u> <u>include/patterns/EntityCreator.h:14</u> <u>include/patterns/WorldObjectCreator.h:13</u> <u>src/patterns/EntityFactory.cpp:19</u> <u>src/patterns/EntityFactory.cpp:19</u>

Diagram relation/contract	Current evidence
EventBus, Observer, and the RAI token	include/patterns/EventBus.h:36 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/EventBus.h#L36) include/patterns/IObserver.h:17 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/IObserver.h#L17) include/patterns/Subscription.h:23 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/patterns/Subscription.h#L23) src/patterns/EventBus.cpp:17 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L17) src/patterns/EventBus.cpp:51 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L51) src/patterns/EventBus.cpp:196 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L196) src/patterns/EventBus.cpp:210 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L210) src/patterns/EventBus.cpp:242 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L242) src/patterns/EventBus.cpp:242 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/patterns/EventBus.cpp#L242)

Diagram relation/contract	Current evidence
SoundManager Singleton observer and subscription seam	include/core/SoundManager.h:116 (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SoundManager.h:127) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/include/core/SoundManager.h:251) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp:40) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp:45) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp:48) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp:105) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/SoundManager.cpp:109) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp:62) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp:65) (https://github.com/baoduong2342007/CS202-Group04-FinalProject/blob/main/05_Source_Code/src/core/Game.cpp:68)

Diagram relation/contract	Current evidence
Command ownership and dispatch	<u>include/patterns/InputHandler.h:40</u> <u>include/patterns/InputHandler.h:86</u> <u>include/patterns/ICommand.h:16</u> <u>src/patterns/InputHandler.cpp:18</u> <u>src/patterns/InputHandler.cpp:46</u> <u>src/patterns/InputHandler.cpp:62</u> <u>src/states/PlayState.cpp:62</u> <u>src/states/PlayState.cpp:62</u>

The diagrams do not attempt to list every private field, every tile object, or every gameplay enum. What is omitted is implementation detail that does not change the ownership/contract boundaries drawn; a class not appearing must not be inferred to be non-existent. A Mermaid parser/renderer is not installed in the workspace, so the document's local verification relies on standard `classDiagram` blocks, class IDs free of dangerous namespace/generic characters, and delimiter balance checks.